

COURSE SYLLABUS

CSC16111 – Topics in Computer Vision

1. GENERAL INFORMATION

Course name:	Topics in Computer Vision
Course name (in Vietnamese):	Chuyên đề thị giác máy tính
Course ID:	CSC16111
Knowledge block:	Graduation
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	No
Prior-course:	Computer Programming, Digital Image Processing, Computer Vision
Instructors:	

2. COURSE DESCRIPTION

This course aims to develop the ability to study the frontiers of computer vision researchers. This course will cover advanced topics in computer vision, such as 1) Image Segmentation, 2) Object Detection, 3 Video Summarization, 4) Person Re-Identification, 5) Instance Segmentation. Students will work on group projects related to concrete research problems and present their research results in the form of seminars. There are many open problems in this area suitable for investigation by Master's or PhD students.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
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G1	Work on an individual level and collaborate in groups to present a number of technical reports in the field of computer vision.	2.2, 2.3.1
G2	Describe and analyze the main research challenges in the field of computer vision.	2.4.3, 2.4.5
G3	Summarize research literature and state-of-the-art techniques for solving the challenging research problems in those areas.	2.4.3, 2.4.5, 2.5.1, 4.1
G4	Model and formulate problems, propose effective solutions to the problem and implement algorithms using suitable programming languages.	4.2, 5.1.2, 5.4.2
G5	Analyze the results and effectively evaluate the results on benchmark datasets.	4.2, 4.3
G6	Write a report and research paper according to journal or conference templates (IEEE, Springer, Elsevier)	5.3

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Establish, organize, operate and manage groups	U
G1.2	Participate in group discussions and argue on subject topics	U
G1.3	Analyze, synthesize and write technical documents according to a given template for individual or group collaboration	I, T
G2.1	Know and understand specialized English terms of the subject	I
G2.2	Read and understand English materials related to lectures.	I
G3.1	Understand the models in research papers in specific domain	I, T, U
G3.2	Analyze the state of the art methods to resolve the problem in specific topics	I, T
G4.1	Recognize the problem and design the model for implementation phase	I, I, U

G4.2	Evaluate and implement the models using suitable frameworks and programming languages	I, U
G5.1	Design and construct a dataset to test and proof the solution in specific domain	I, U
G5.2	Analyze and evaluate the results on datasets in experiments	I, T, U
G6.1	Write & present a result of research topic that use journal or journal or conference templates (IEEE, Springer, Elsevier)	I, T, U

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Introduce challenges and problem in Computer Vision	G1.1, G1.2, G1.3, G2.1, G3.1	Lecturing Q&A, Group discussion	
2	Image Segmentation	G2.1, G3.1, G6.1,	Reading, Video Demonstration, Q&A	QZ1: Quiz 1
3	Object Detection	G6.1	Lecturing Demonstration, Discussion, Q&A	HW1
4	Video summarization	G6.1	Reading, Video Demonstration, Discussion, Q&A	QZ2: Quiz 2
5	Person Re-Identification	G6.1	Lecturing Demonstration,,Q&A	HW2
6	Instance Segmentation	G6.1	Reading, Video, Question & answer	QZ1: Quiz 3



			Case study and Discussion	
7	Project Representation	G4.1, G5.1, G5.2, G6.1	Seminar, Q&A, Discussion	HW3, HW4: Post-harvest
8	Review	G2, G4.1, G5.2	Lecturing Q&A, Discussion Project submitted	

LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Introduce the outline of content, programming environment for final project	G4.2, G5.1, G5.2, G6.1	Explain and demonstrate Q&A	QZ1
2	Answer and explain the question for final project	G1, G3, G4	Q&A Case study and discussion	HW3, and HW4

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			20%
A1 1	Quizzes: QZ1, QZ2, and QZ3	Small quizzes in class for each topic	G5.1, G2.1, G2.2, G3.1, G3.2	10%

A1 2	Homework: HW1, HW2, HW3, and HW4	HW1, HW3: reading comprehension and writing reports in English HW2, HW3, HW4: practicing based on knowledge taught in class	G5.2, G2.1, G2.2, G4.1, G4.2, G3.1, G3.2	10%
A2	Project			40%
A2 1	Research & represent a subject in classes topics		G3, G4, G5, G6	40%
A3	Final Exam	Open book exam. Describe the understanding of different topics, analyze & program to solve problems	G3	40%

7. RESOURCES

Textbooks

- No textbook

Conferences:

- International Conference on Computer Vision (ICCV)
- Computer Vision and Pattern Recognition (CVPR)
- International Conference of Pattern Recognition (ICPR)

Others

- [Deep Learning, Ian Goodfellow and others](#)
- [Neural Networks and Deep Learning, Michael Nielsen](#)
- OpenCV, Tensorflow, PyTouch

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss the topics. However, individual work must be done and submitted on your own.